STA 2002, Summer 2024
Probability and Statistics II

Introduction

Zhenxing Guo

Acknowledgement to Prof. Liyan Xie

Outline

- Course organization
- Course introduction
- Review of STA2001

Course staff

- **Instructor:**
  - *Zhenxing Guo*
    - **Office:** DY 320a
    - **Office Hours:** Friday 2pm – 3pm (Zoom available by requested)
    - **Email:** guozhenxing@cuhk.edu.cn
- **Lectures**:
    - Room: Teaching B Bldg 104
    - Times: Wen&Thu 1:30-3:20pm，Fri 10:30am-12:20pm
    - Zoom ID: 319 926 3469
    - Zoom Passcode: 282178
- **TA**
  - *Fanyi Peng*
    - **Office:** txc604
    - **Office Hours:** Mon 3pm – 4pm (Zoom available by requested)
    - **Email:** 223040234@link.cuhk.edu.cn

Tutorials

- **Tutorials start on the second week, the week of June.12nd**
- **All materials for tutorials will be uploaded to BB.**

Assignment

- **Five assignments in total**
- **Electronic submission on blackboard (https://bb.cuhk.edu.cn)**
- **No late homework will be accepted**

Learning Assessment Strategy

Component/ method	% weight
Assignments	25
Mid-Term Exam	25
Final Exam	50

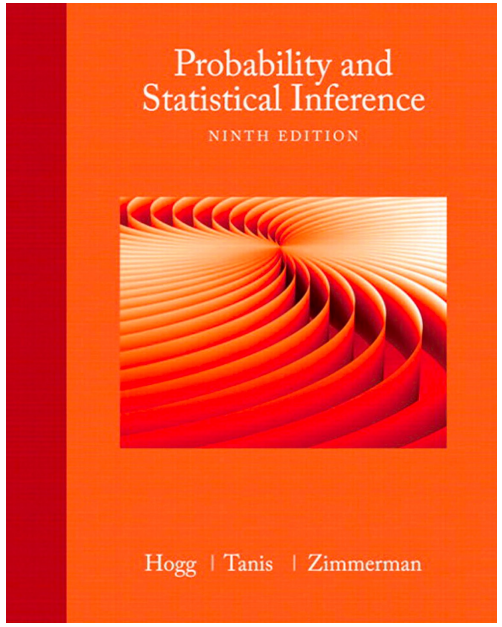
- **Midterm (tentative): June 30 (Sunday) 2:30 – 4:30 PM. Let the instructors know immediately if this timeslot crashes with other course(s).**
- **There will be no make-up exams**
- **Zero tolerance policy when it comes to cheating so, DON'T CHEAT**
- **Missing the midterm or final exam without prior notification to and approval of the instructors will automatically result in the "0" grade for the exam.**

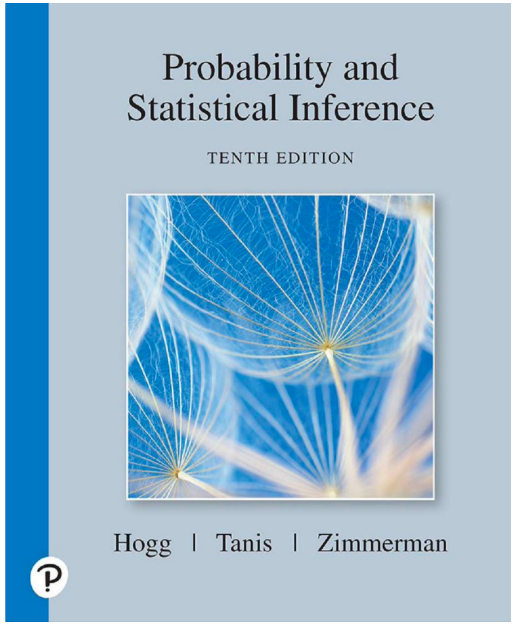
Resources

- **Textbook:** Hogg, R. V., Tanis, E. A. Zimmerman, D. L. (2015) Probability and Statistical Inference, 9th/10th edition, Pearson

  (Chapter 6-9)

- **Blackboard site:** https://bb.cuhk.edu.cn





Tentative Calendar

Week	Content/ topic/ activity
1	General Introduction. Review of probability basics and families of distributions. Descriptive statistics.
2	Point estimation (Maximum Likelihood Estimation and Method of moments). Confidence interval (CI) estimation for means; CI for the difference of two means; CI for proportions. Sample size.
3	Tests of statistical hypotheses, Critical region, Significance level, p-value, Power, t-test, F-test. Tests about proportions.
4	More tests (Sample size; Power of a statistical test). Review Session. **Midterm Exam**.
5	Distribution-free CIs and tests. Chi-square goodness of fit. Contingency tables. ANOVA.
6	Simple Linear Regression. More tests concerning regression and correlation.
7	Likelihood ratio tests. Bayesian estimation. Final Review.

Outline

- Course organization
- Course introduction
- Review of STA2001

History of Statistics

HISTORICAL NOTE

The earliest statistical recordings include:

- ancient **Babylonians** recorded their crop yields on clay tablets.
- ancient **Egyptian pharaohs** recorded their wealth on stone walls.

More recently:

- **William Playfair** (1759-1823) developed the histogram to display data.
- **Florence Nightingale** (1820-1910) kept records of the dead and injured during the Crimean War. She developed and used graphs extensively.
- **John Tukey** (1915-2000) invented the stemplot in 1972 and the boxplot in 1977.





History of Statistics

- ✓ The History of statistics can be said to start around 1749 although
- ✓ The modern field of statistics only emerged in the late 19th and early 20th century in three stages



Karl Pearson, the founder of mathematical statistics

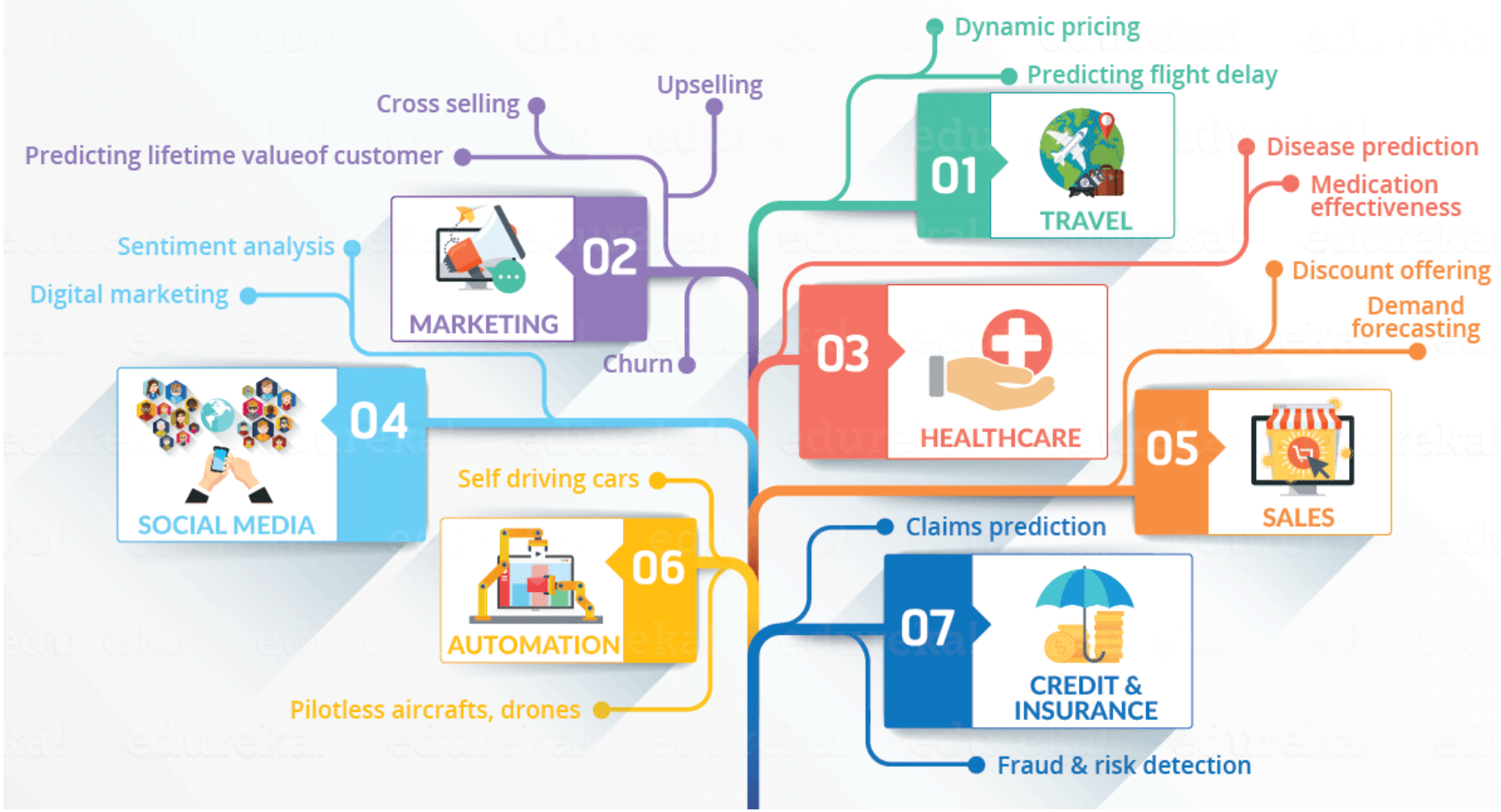


Ronald Fisher, “a genius who almost single-handedly created the foundations for modern statistical science”



Pierre-Simon, marquis de Laplace, one of the main early developers of Bayesian statistics

Statistics in real world



Course Contents

- ✓ Applications of laws of large numbers and central limit theorem in parameter estimation and test statistic construction.
- ✓ Point estimation, maximum likelihood estimation, confidence intervals.
- ✓ Hypothesis testing: basic concepts and more advanced techniques.
- ✓ Regression analysis, contingency tables, analysis of variance.
- ✓ Bayesian estimation (if time permits)

Basic Concepts in Statistics

Statistics and statistical thinking

The field of Statistics deals with the *collection*, *presentation*, *analysis*, and *use* of data to **model** systems, make decisions, solve problems, and design products and processes

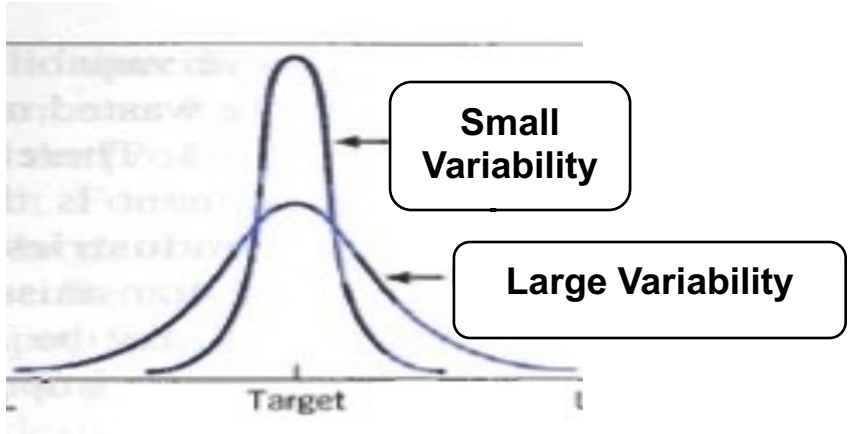
Statistics is the science of data

Examples: Statistics helps us

- Predict the demand of a product / the stock prices
- Select the best supplier with the least lead time (or highest quality)
- Monitor and control a process
- Simulate and model an ER
- Determine the probabilistic distribution of machines life
- Design new products
- Understand customer preferences

Variability in data

- **Samples are random**
  - **individual variability**
  - **noise**
  - **measurement errors**
  - **the world has randomness**

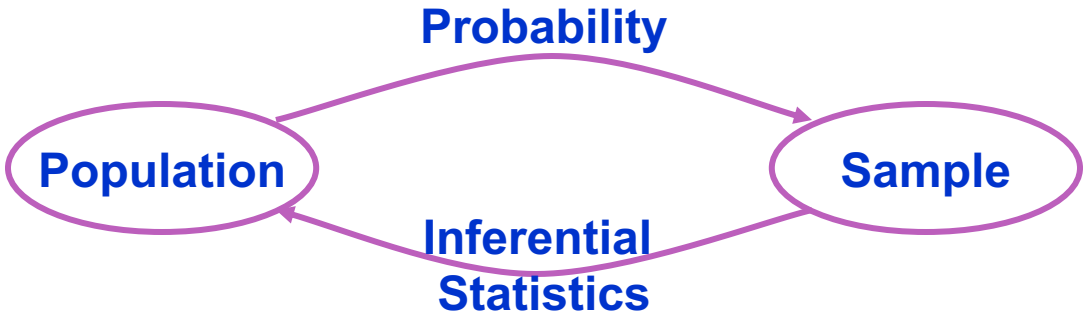


- **Random is the same as variability:**

successive observations of a system or phenomenon do *not* produce exactly the same result.

- **We capture the randomness by probability models**

Probability Vs. Statistics





Probability: given the information in the pail, what is in your hand?

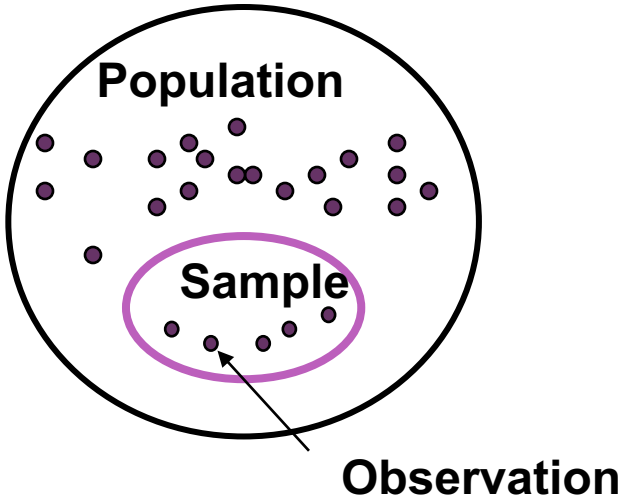


Statistics: given the information in your hand, what is in the pail?

Population Vs. Sample

- **Population:** a finite well-defined group of ALL objects which, although possibly large, can be enumerated in theory (e.g. investigating ALL the bearings manufactured today).

- **Sample:** A sample is a SUBSET of a population



Estimation

Hospital waiting time:

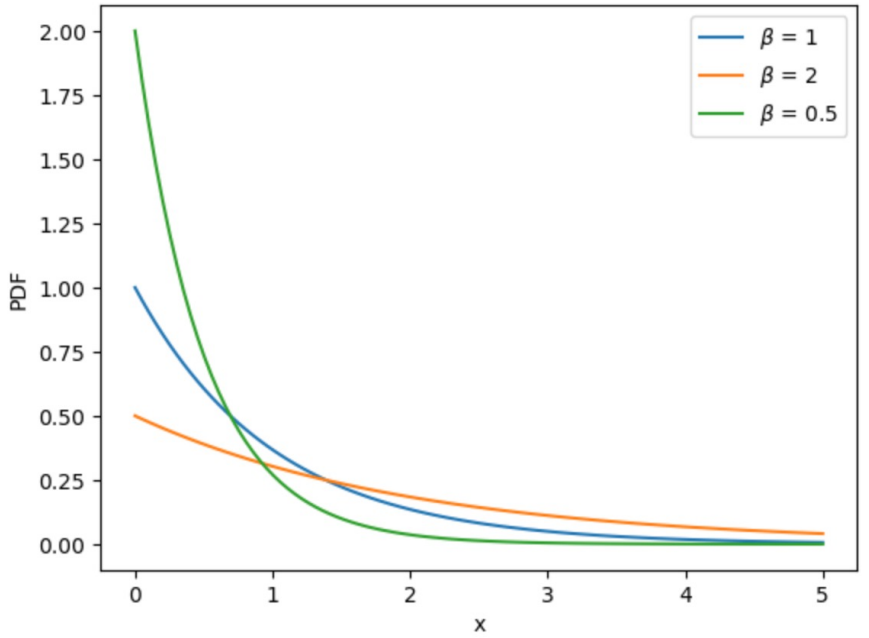
4.80	4.92	5.08	4.90	4.98	5.14	5.02	5.07	5.05	4.95
4.74	5.09	5.01	5.07	4.93	5.05	5.09	4.89	5.15	5.01
5.31	5.42	5.25	5.35	5.22	5.39	5.35	5.33	5.22	5.32
4.97	5.13	4.98	5.17	4.87	5.09	4.77	5.12	5.17	5.09
5.07	5.00	5.02	4.97	4.88	5.08	5.08	4.98	4.99	4.93

✓Determine a probability distribution of a population based on a random sample
✓Estimate parameters of a distribution based on a random sample

$$f(x;\beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta} & x \geq 0, \\ 0 & x < 0. \end{cases}$$

True parameter value

$$\widehat{\boldsymbol{\beta}} = \mathbf{5.07}$$

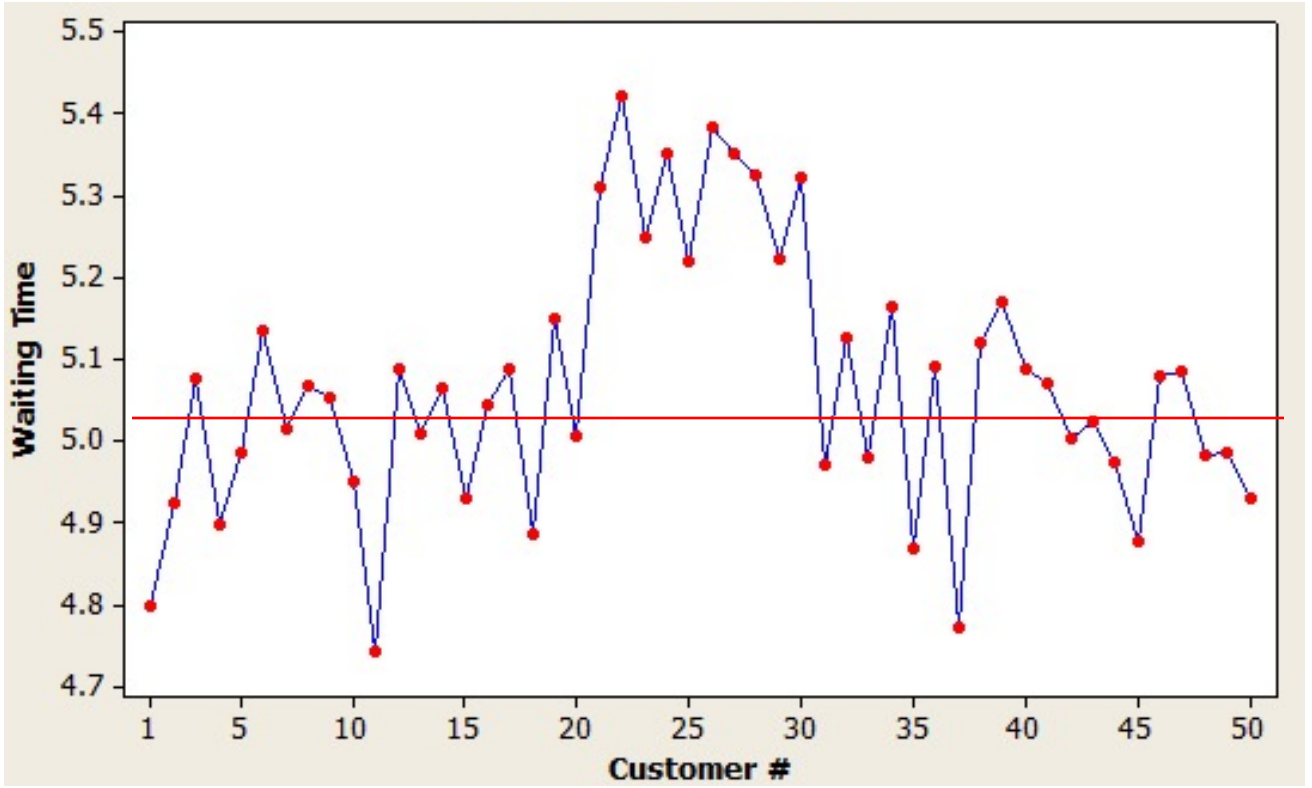


Confidence interval

How confident we are given the variability of data?

✓Construct confidence intervals for parameters of a distribution

5.07



$$\beta \in [5.07 - 0.16. 5.07 + 0.16]$$

Hypothesis test

Given the average wait time 5.07

Null hypothesis $\beta \leq 5$

Alternative hypothesis $\beta > 5$

Which one is true?

Data

↓

Statistics

↓

Decision

Example: Comparing two populations

- **Which drug is more effective**





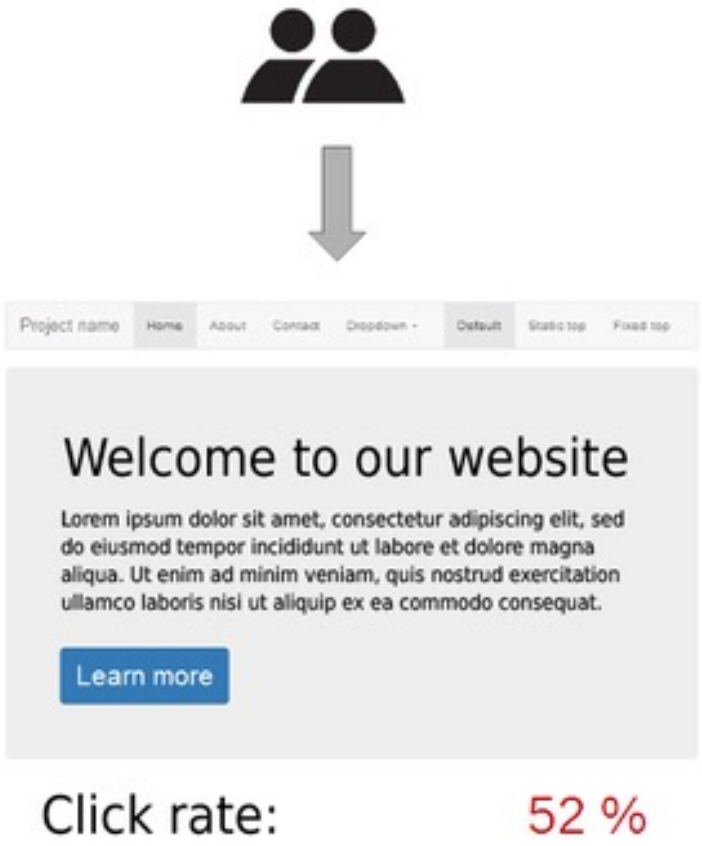


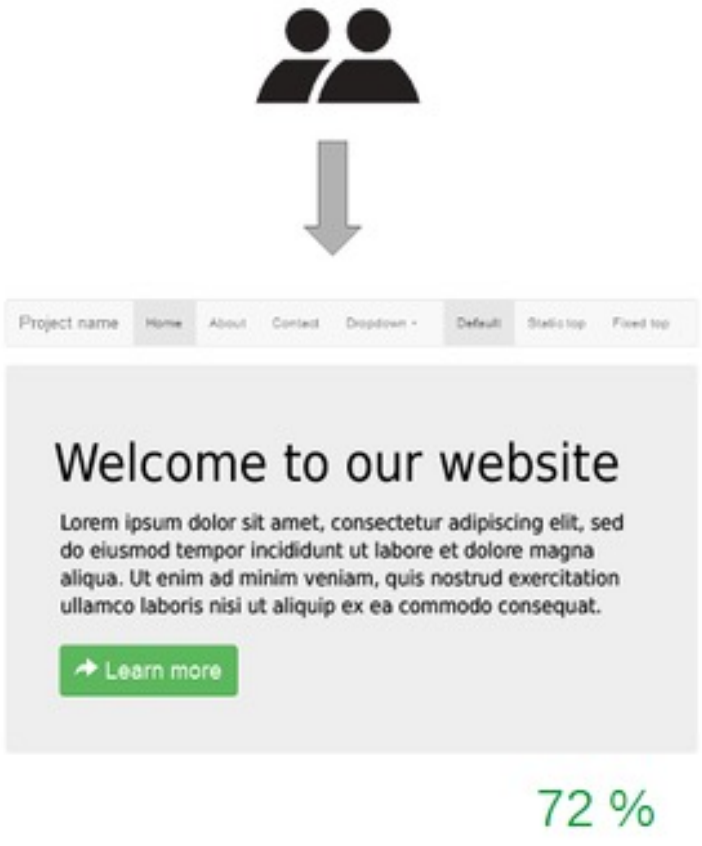


Group 1		Group 2	
16.03	16.01	16.02	16.03
16.04	15.96	15.97	16.04
16.05	15.98	15.96	16.02
16.05	16.02	16.01	16.01
16.02	15.99	15.99	16.00

Example: Internet company

A/B testing





Example: Coke Cola



The Coca-Cola Co. is releasing its new Georgia Peach and California Raspberry flavors.

THE COCA-COLA CO.

Regression and variable selection

- ✓ Predict a response variable based on one or more predictor variables
- ✓ Identify important factors influencing a response variable

What are important variables affecting the waiting time in hospital waiting room?

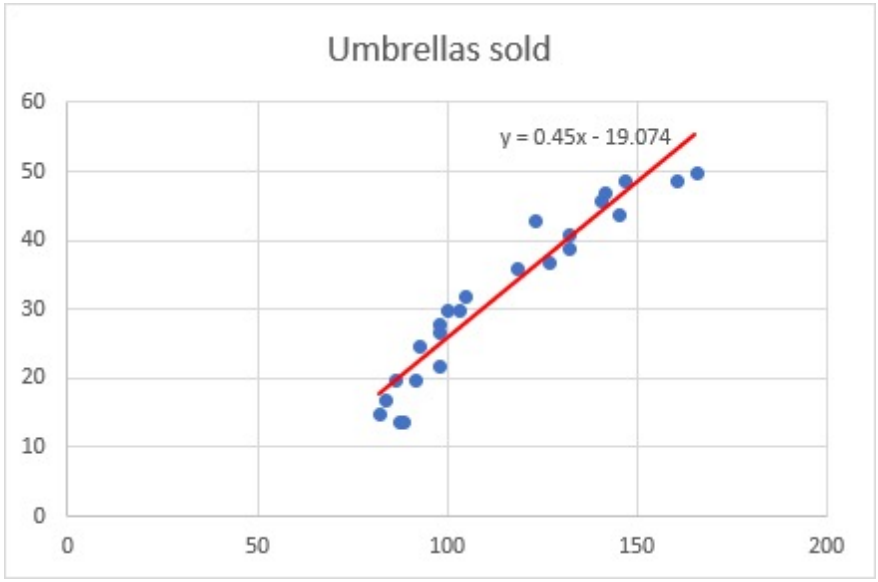
Type of emergency cares
Physicians' experience and skills
Time of day
Number of nurses
Capacity of waiting room

→ **Waiting time in hospital**

Linear regression

✓ Predict a response variable based on one or more predictor variables

	A	B	C
1	**Month**	**Rainfall (mm)**	**Umbrellas sold**
2	Jan	82	15
3	Feb	92.5	25
4	Mar	83.2	17
5	Apr	97.7	28
6	May	131.9	41
7	Jun	141.3	47
8	Jul	165.4	50
9	Aug	140	46
10	Sep	126.7	37



Outline

- Course organization
- Course introduction
- Review of STA2001

Prerequisites: Review Questions

Prerequisite: STA2001 (or Textbook Chapter 1-5)

How many "Yes" do you get?

1. Do you know probability distribution, pdf, CDF?
2. Do you know difference between discrete and continuous random variables?

3. Are you familiar with moments?
4. Do you know how to calculate mean/variance of a random variable?

5. Do you know Exponential distribution and Gamma distribution?
6. Are you familiar with Binomial distribution?
7. Are you familiar with Poisson distribution?
8. Do you know how to calculate probability associated with a Normal distribution?

9. Can you calculate the integral and derivative of a function?
10. Do you know how to calculate the inverse of a 2-by-2 matrix and the product of any two matrices by hand?

Discrete distributions

- ***Bernoulli(p)***
- ***Binomial(n,p)***
- ***Negative binomial(r,p)***
- ***Discrete uniform(m)***
- ***Geometric(p)***
- ***Poisson(λ)***
- ***Hypergeometric(N,N_1,N_2)***
- **……**

Continuous distributions

- ***Normal(μ,σ^2)***
- ***Exponential(θ)***
- ***Gamma(α,θ)***
- ***Chi-square(r)***
- ***Student's t-distribution(r)***
- ***F-distribution(r_1,r_2)***
- ***Uniform(a,b)***
- **……**

Summary

- Statistics is the science of data
- To study data using statistics, we will need
  - tools and concepts
    - Population vs. samples
    - Probability vs. statistics
    - Model
  - Programming: R, Python, Matlab, ...
- Major topics
  - Estimation
  - Confidence interval
  - Hypothesis test
  - Linear regression
  - Bayesian

Qualitative

Quantitative